

Name 1

Datum:

Name 2

Platz-Nr:

41 - Beta Decay

1 Characteristics of the Counter Tube

V											
t											
$N(V)$											

 $V_{\text{ideal}} = \dots\dots\dots V$ Voltage dependence of the count rate N at V_{ideal} :The smaller $\frac{\Delta N}{\Delta V}$ the $\dots\dots\dots$ (better / worse) is the counter tube.

2 Activity of the Source

- $t = \dots\dots\dots$ (measuring time)
- $N = \dots\dots\dots$ (incident particles in time t)
- $N_{\text{BG}} = \dots\dots\dots$ (background in time t , measured with 8 mm brass absorber)
- $N_{\text{eff}} = N - N_{\text{BG}} = \dots\dots\dots$ (effective number of incident particles in time t)

Calculation of the solid angle:

Diameter of the aperture $d = \dots\dots\dots$ mmHalf the angle of the cone $\theta = \dots\dots\dots$ degAnd hence the **acceptance** is $\varepsilon = \dots\dots\dots$

What is the solid angle? Note that the source emits approximately uniformly in all directions. However, the detector “sees” only a fraction of this, the so-called solid angle fraction (referred to as “acceptance”). The calculation of the solid angle should be learned in advance.

Activity of the current source = $\dots\dots\dots$ BqActivity of the current source when neglecting the protective steel foil = $\dots\dots\dots$ Bq

3 Calculations relevant for radiation protection

- $1/e$ time of Sr: $\dots\dots\dots$ years.
- $1/e$ time of Y: $\dots\dots\dots$ hours.
- Assuming the source was manufactured on 1/1/1950: What was the activity at the time of manufacture? $\Rightarrow$
The activity was $\dots\dots\dots$ Bq.
- The activity of the source decreases to 10 kBq in $\dots\dots\dots$ years.
- How long does it take for the average annual dose in Switzerland (= 5.5 mSv) to be reached if the complete radiation of the source contaminates 1 cm^2 of skin?
 $\Rightarrow \dots\dots\dots$

4 Absorption and β -Energy

Hint: Density $\rho(\text{Al}) = 2.69 \text{ g cm}^{-3}$

only for students of D-PHYS

x [mm]	$\rho(\text{Al})x$ [gcm ⁻²]	t [sec]	N	N_{eff}	measured activity [Bq]	$E(x)$ [MeV]	$\frac{E_{\text{max}}}{E(x)}$ – [MeV]
0.00							
0.25							
0.50							
1.00							
1.25							
1.50							
1.75							
2.00							
2.25							
2.50							
2.75							
3.00							
3.50							
4.00							

$x_{\text{max}} = \dots\dots\dots$ mm, and hence $E_{\text{max}} = \dots\dots\dots$ MeV

Energy loss in the steel foil = $\dots\dots\dots$ keV

$E_{\text{max, source}} = \dots\dots\dots$ MeV

$\mu = \dots\dots\dots \text{ cm}^{-1}$, and hence E_{max} (for $\mu = \dots\dots\dots \text{ cm}^{-1}$) $\dots\dots\dots$ MeV

$E_{\text{max Quelle}} (\mu) = \dots\dots\dots$ MeV

5 E_{max} for students of D-PHYS

Coefficient $n = \dots\dots\dots$, and thus $E_{\text{max}} = \dots\dots\dots$ MeV

$E_{\text{max, source}} = \dots\dots\dots$ MeV